End-Semester Examination December 2024

Name of the Program: B.Tech | *Semester: 7th*
Name of the Course: Bio-medical Electronics | *Course Code: TOE-703*

Time: 3 Hrs | Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub-questions among a, b & c in each main question
(iii)Total marks in each main question are twenty.
(iv)Each question carries 10 marks

Q1	(20marks)	CO5
(a)	How do X-ray and MRI imaging systems differ in their diagnostic capabilities?	
(b)	Describe the role of computed tomography (CAT scan) in medical imaging	
(c)	Explain the principles and applications of ultrasonic diagnosis and eco-cardiography.	
Q2		CO4
(a)	Explain how the physiology of the respiratory system influences the design of respiratory measurement devices.	
(b)	What are the principles and applications of spirometers in clinical settings?	
(c)	Discuss the role of inhalators, ventilators, and nebulizers in respiratory therapy.	
Q3		CO1
(a)	Describe the various types of bioelectric signal amplifiers used in medical devices. Explain the role of filters in biomedical signal processing.	
(b)	Explain the role of filters in biomedical signal processing.	
(c)	Describe the key factors influencing the design of biosensors.	
Q4		CO2
(a)	What are the differences between active and passive transducers? Provide examples specific to biomedical applications.	
(b)	Explain the applications of biomedical equipment in monitoring and diagnosis.	
(c)	Explain the working principles of body surface electrodes, needle electrodes, and biochemical transducers.	
Q5		CO3
(a)	Explain the terms ECG and EEG in detail.	
(b)	Explain the working of Holter recording and its role in continuous ECG monitoring.	

(c)	What are the techniques used to measure blood flow and heart sounds.	